Experiment 1: Selection Sort on Random, Reverse and Sorted Arrays

1.Aim

To implement Selection Sort in C and analyze its performance on random, reverse-sorted, and already-sorted arrays.

2. Theory

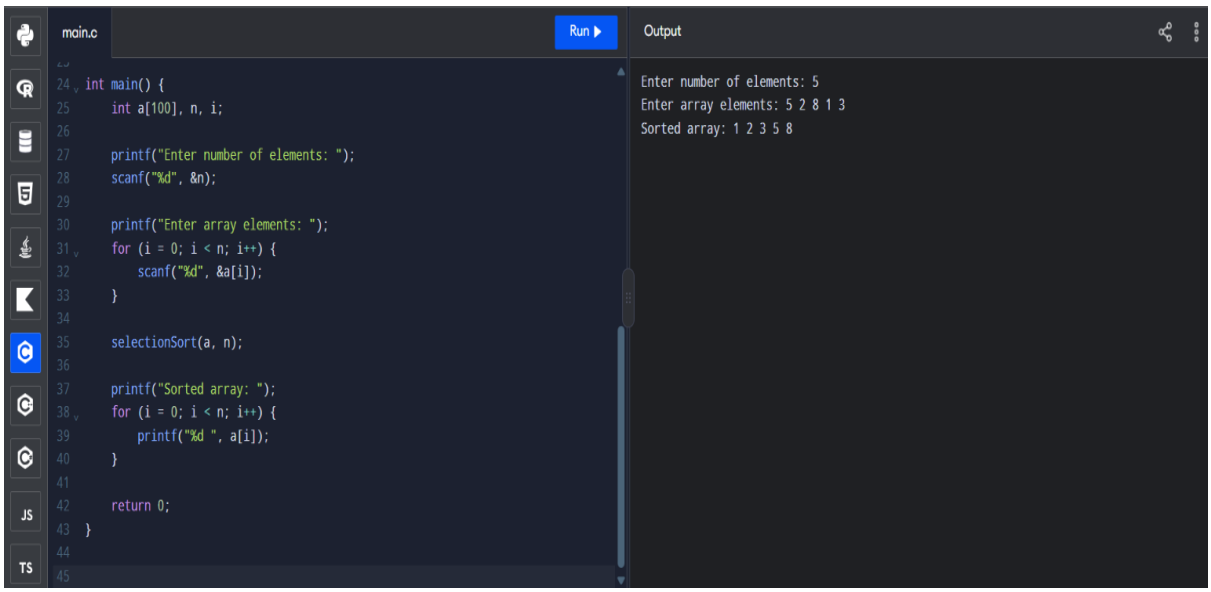
Selection Sort is a sorting algorithm that repeatedly finds the smallest element from the unsorted part of an array and places it in the correct position.

The array is divided into two parts: the sorted portion and the unsorted portion. During each pass, the minimum element from the unsorted portion is selected and swapped with the first element of that portion.

Selection Sort can be tested using three types of input:

- Random array: Elements are arranged in an arbitrary order.
- Reverse-sorted array: Elements are arranged in descending order.
- Sorted array: Elements are already arranged in ascending order.

Selection Sort performs approximately the same number of comparisons for all three input types.



5. Complexity Analysis

Case	Time complexity
Best case	\(O(n^2)\)
Average case	\(O(n^2)\)
Worst case	\(O(n^2)\)
Auxiliary space	\(O(1)\)

Result: Selection Sort successfully sorts random, reverse-sorted, and sorted arrays in ascending order.